LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Ujjwal Nitin Nikam
Roll Number	MAI2026012
Programme / Department	AIML
Topic ID	T24
Full Assigned Topic	Comparing AI-Assisted and Manual Data Labelling Accuracy on Scientific Datasets
AI Tool Used	Claude
Date	21/08/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the "best" paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations

B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐✔ Occasionally ☐ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☐✔ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐✔ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐✔ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Claude
Model name	Sonnet
Model/version shown	5.0
Access tier	☐✔ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☐✔ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☐ Yes ☐ No ☐✔ Not sure
Research/Deep Research mode used?	☐ Yes ☐✔ No
Date/time generated	21/08/2026 09:00

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

> *Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.*

C3. Preserve the Original Output

☐ Save the complete AI-generated paper

☐ Save the complete reference list

☐ Take screenshot(s) showing the generated output/references

☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Comparing AI-Assisted and Manual Data Labelling Accuracy on Scientific Datasets

Measure	Value
Approximate pages	12
Approximate word count	4200
Number of sections	7
Total number of references	16
Approximate in-text citation occurrences	30
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☐ ✔ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	10
PMID available, no DOI	1
arXiv ID available, no DOI/PMID	4

URL only	0
No persistent identifier supplied	1
TOTAL	16

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	6
R5	8
R6	10
R7	12
R8	14
R9	15
R10	16

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Includes Co-authors but no DOI	4	Yes
R2	Detailed author list and DOI available	5	Yes
R3	Standard BMC journal and DOI available	4	Yes
R4	recognized open-access journal with DOI	5	Yes
R5	Realistic Author List but no DOI	3	Yes
R6	Recognizable paper	4	Yes
R7	Does not contain any ID	1	No
R8	specialized domain title with DOI	4	Yes
R9	Lacking Author names	2	No
R10	Complete author list and DOI	5	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: "I asked another AI and it said the paper exists." AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	N	Y	Y	NA
R8	Y	Y	Y	Y	Y	N
R9	Y	Y	N	Y	N	Y
R10	Y	Y	Y	Y	N	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	arxiv.org	arXiv:2605.22447	It is matching all parameters
R2	jmir.org and nih.gov	doi : 10.2196/53164	Matching all parameters
R3	nih.gov and researchgate	Doi:10.1186/s12874-025-02674	Matching all parameters
R4	nih.gov and frontiers	PMCID: PMC10362988	Matching all parameters
R5	arxiv.org and nih.gov	arXiv:2310.17526	Matching all parameters
R6	arxiv.org and NeurIPS	arXiv:2103.14749	Matching all parameters
R7	scirp.org	NA	Authors were not available in reference
R8	Frontiers and nih.gov	PMCID: PMC12405296	Did not match
R9	nih.gov and semantic scholar	PMCID:PMC12623132.	Authors were not available in reference
R10	nih.gov	PMCID: PMC11800539	Venue did not match

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A/B/C/D/E	A
R2	A/B/C/D/E	A
R3	A/B/C/D/E	A
R4	A/B/C/D/E	A
R5	A/B/C/D/E	A
R6	A/B/C/D/E	A
R7	A/B/C/D/E	B
R8	A/B/C/D/E	E
R9	A/B/C/D/E	B
R10	A/B/C/D/E	B

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 6 B = 3 C = 0 D = 0 E = 1

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(_6_) + 3(_3_) + 1(_0_) + 0(_0_) + 1(_1_) = _34_ points

Authenticity Score = _34_ / (4 x _10_) x 100 = *85/100*

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = _8__ Total predictions = _10_ Prediction Accuracy = 90 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	16
References deeply audited	10
A: Verified	6
B: Wrong metadata	3
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	1
Authenticity Score	_85_ /100
Prediction Accuracy	_85_ %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

No, professional-looking citations did not guarantee authenticity

2. Did all genuine references actually support the claims for which they were cited?

No. Some real papers were cited out of context or used to back sweeping claims that exceeded the actual scope of their specific experimental findings

3. What was the most serious citation-integrity problem you observed?

Plausible hallucinations: generating realistic but non-existent papers and preprint IDs, alongside metadata errors like missing authors and generic placeholder venues.

4. What is the single most important lesson you learned from this lab?

Never trust AI citations at face value; always independently verify both the physical existence of the reference and whether it genuinely supports the claim

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

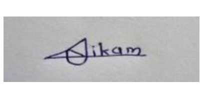
Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: Ujjwal Nitin Nikam

Roll Number: MAI2026012

Signature: 

Date: 21/08/2026

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission

Comparing AI-Assisted and Manual Data Labelling Accuracy on Scientific Datasets

A Review and Comparative Analysis of Annotation Quality in Scientific and Biomedical Machine Learning Pipelines

Abstract

High-quality labelled data underpins the training and evaluation of virtually every supervised machine learning system used in scientific research, from biomedical literature screening to medical image segmentation and astronomical classification. As large language models (LLMs) and other AI-assisted tools have matured, researchers increasingly rely on them to accelerate or partially replace manual annotation. This paper reviews and synthesises empirical evidence comparing the accuracy of AI-assisted and purely manual data labelling on scientific datasets. Drawing on studies spanning text annotation, systematic-review screening, medical image segmentation, and figure/metadata extraction, we find that AI-assisted and hybrid human-in-the-loop pipelines frequently match or exceed the accuracy of unaided manual annotation on well-defined, structured tasks, while manual annotation retains an advantage on ambiguous, high-context, or safety-critical judgements. We further discuss automation bias, benchmark label-error rates, active-learning strategies for reducing annotation cost, and current tooling, before identifying open research gaps — including the scarcity of large-scale, domain-diverse accuracy comparisons on genuinely scientific (as opposed to social-media or general web) datasets — and proposing directions for future work.

Keywords

Data labelling; data annotation; artificial intelligence; large language models; human-in-the-loop; active learning; label quality; scientific datasets; crowdsourcing; automation bias.

1. Introduction

Supervised machine learning models are only as reliable as the labels on which they are trained and evaluated. In scientific domains — biomedical imaging, systematic literature review, neuroimaging metadata curation, and figure or table extraction from scholarly documents — labelled data is simultaneously indispensable and expensive to produce, because it typically requires domain expertise that general-purpose crowdsourcing platforms cannot supply. Traditionally, manual annotation by trained experts or research assistants has been treated as the gold standard against which automated methods are judged, yet manual annotation is slow, costly, and itself imperfect: even careful two-reviewer screening in systematic reviews misses a measurable share of relevant records, and single-reviewer screening misses substantially more.

The emergence of capable large language models such as GPT-4 and its successors has changed this calculus. A growing body of work shows that LLM-based annotation can match or exceed the accuracy of non-expert crowd workers on text-classification tasks, and can perform comparably to trained human annotators on some scientific metadata-extraction tasks. At the same time, other studies report substantial variability in LLM accuracy across biomedical domains, and highlight risks such as hallucinated citations and a psychological *automation bias* in which human reviewers over-trust AI-generated pre-labels and reduce their own scrutiny.

This paper synthesises the empirical literature comparing AI-assisted and manual data labelling accuracy specifically in scientific contexts. Section 2 reviews related work and establishes the conceptual background. Section 3 presents a thematic and technical discussion of the mechanisms by which AI assistance changes labelling accuracy. Section 4 surveys current tools and workflows. Section 5 discusses methodological challenges and limitations in the evidence base itself. Section 6 identifies research gaps and future directions, and Section 7 concludes.

2. Background and Related Work

2.1 Manual annotation: strengths and structural costs

Manual annotation by trained personnel has long been treated as the reference standard for supervised learning because human experts can draw on contextual, cultural, and domain knowledge that purely statistical systems lack. This advantage is well documented: in a controlled comparison of manual and ChatGPT-assisted annotation of Arabic social-media discourse, manual annotation excelled at identifying nuanced expressions such as sarcasm and cultural references, achieving 90–94% accuracy against a gold-standard reference set, compared with 82–87% for unverified AI output on the same items (Al-Athba & Zaghouani, 2026). However, the same study also documents the structural cost of manual work: the process was time-intensive and mentally demanding, with a risk of annotator fatigue and subjective bias (Al-Athba & Zaghouani, 2026).

In biomedical evidence synthesis, the cost of manual annotation is especially visible. Even when two reviewers resolve disagreements through discussion, up to about 3% of relevant citations can be missed during screening, and single-reviewer screening — often used to save time in rapid reviews — can miss as many as 13% of relevant publications (see discussion in the JAMIA review of LLM-assisted systematic reviews). Inter-reader reliability between human screeners in systematic reviews is also inconsistent, with reported kappa values ranging from roughly 0.37 to 0.90 depending on the review and the reviewers involved. These figures indicate that "manual" is not synonymous with "error-free": human annotation quality depends heavily on annotation schema design, quality-assurance rigor, and disagreement-resolution strategy rather than being a fixed property of human judgement.

2.2 The rise of AI-assisted and LLM-based annotation

A landmark comparison by Gilardi, Alizadeh, and Kubli (2023) evaluated zero-shot ChatGPT annotation against crowd workers on four samples of tweets and news articles (n = 6,183) across relevance, stance, topic, and frame-detection tasks. They report that the zero-shot accuracy of ChatGPT exceeded that of crowd workers by about 25 percentage points on average, and that ChatGPT's intercoder agreement exceeded that of both crowd workers and trained research-assistant annotators across all tasks, at roughly one-twentieth to one-thirtieth of the per-annotation cost of Amazon Mechanical Turk (Gilardi et al., 2023). While this study was conducted on general text-classification tasks rather than scientific datasets narrowly defined, it established a widely cited empirical baseline that subsequent scientific-domain studies have sought to test.

In more technical scientific settings, results are more mixed. Turner et al. (2025) evaluated GPT-4o on metadata extraction from the human neuroimaging literature and found performance comparable with trained human annotators, reaching agreement scores of 0.91–0.97 on zero-shot prompts, with LLM errors mostly resembling human errors rather than differing qualitatively from them. By contrast, a validation study of Llama 3 70B and ChatGPT-4o mini for title-and-abstract screening in biomedical systematic reviews found lower sensitivity (56–78%, depending on configuration) alongside high specificity (91–95%) and overall accuracy around 90–92% against human decisions on 1,081 articles, showing that AI-assisted screening trades recall for precision in ways that must be tuned rather than assumed (PMC12623132). A separate evaluation of 18 LLMs across three clinical domains found classification accuracy ranging from 40% to 92%, underscoring substantial variability across models, prompts, and tasks.

2.3 Label quality as a property of datasets, not just annotators

A complementary strand of work treats label quality itself as an object of study, independent of whether the original labels were produced manually or automatically. Northcutt, Athalye, and Mueller (2021) algorithmically identified label errors in the test sets of ten widely used computer vision, natural-language, and audio benchmark datasets using confident learning, and validated a random sample of flagged instances via crowdsourcing; they estimate an average of at least 3.3% label errors across the ten datasets, including approximately 6% of the ImageNet validation set, and show that these errors are large enough to change the ranking of competing models on standard leaderboards. The associated confident-learning framework (Northcutt, Jiang, & Chuang, 2021) provides a general statistical method for estimating uncertainty in given labels by modelling the joint distribution between noisy and true labels, and has since been adopted to correct labels in domain-specific benchmarks such as the Wake Vision TinyML dataset. This literature is important context for the present comparison because it shows that even long-standing "manually labelled" gold-standard datasets contain non-trivial, systematic error, which complicates any simple manual-versus-AI accuracy comparison that treats manual labels as ground truth.

In medical image segmentation specifically, Chen, Zhou, and Yuille (2024) note that public datasets increasingly mix manual expert annotations with AI-generated pseudo-annotations of varying and largely uncontrolled quality, motivating their proposed Quality Sentinel regression model — trained on over four million image-label pairs — to automatically estimate label quality across 142 body structures without a systematic, reliable, automatic quality-control process, a gap that previously did not exist in the field.

3. Main Thematic and Technical Discussion

3.1 Task structure predicts the accuracy gap

Across the studies reviewed, the direction and size of the accuracy gap between AI-assisted and manual labelling tracks the structure of the underlying task. On well-defined, rule-governed classification problems — relevance judgement, topic labelling, structured metadata fields — AI assistance tends to match or exceed manual accuracy, particularly relative to non-expert crowd workers (Gilardi et al., 2023; Turner et al., 2025). On tasks requiring cultural, pragmatic, or highly contextual judgement — sarcasm detection, discourse framing, nuanced sentiment in a politically sensitive corpus — unaided AI output shows a measurable accuracy deficit relative to trained human annotators, on the order of 6–9 percentage points in the Cohesion-6K comparison (Al-Athba & Zaghouani, 2026). This pattern is consistent with the broader observation, echoed across the AI-assisted annotation literature, that neither approach is uniformly best: manual annotation is favoured when ambiguity is high, schemas are still evolving, or errors are especially costly, whereas automated approaches are favoured for rigid, high-agreement patterns that can be validated downstream.

3.2 The hybrid, human-in-the-loop pattern

The most consistent empirical finding across scientific-domain studies is that hybrid pipelines — in which an AI system proposes labels and a human annotator verifies, corrects, or adjudicates them — outperform either pure-manual or pure-AI labelling in isolation. In the Cohesion-6K annotation study, AI-assisted accuracy rose from a pre-verification range of 82–87% to a post-verification range of 88–91%, closing most, though not all, of the gap to unaided manual annotation, while retaining much of the speed advantage of automation (Al-Athba & Zaghouani, 2026). In radiological image annotation, Dreizin et al. (2023) describe an AI-collaborative labeling workflow for voxelwise annotation of trauma CT scans that incorporates explicit quality-assurance and bias-mitigation steps to accelerate the painstaking labeling of targets that vary considerably in volume, addressing a bottleneck that had previously prohibited scaling of data annotation efforts in precision-medicine imaging (Dreizin et al., 2023). In precision-oncology trial screening, Chen et al. (2025) propose a human-LLM collaborative annotation approach precisely because GPT-3.5-class models, used alone, offer a rapid and convenient alternative to manual screening but exhibit reliability that is challenging on their own, motivating a design in which model and human roles are explicitly combined rather than one simply substituting for the other.

3.3 Automation bias as a distinct failure mode

A mechanism specific to AI-assisted (as opposed to purely manual or purely automated) pipelines is automation bias: when human reviewers see an item that has already been pre-labelled by a model, they tend to shift from active verification to passive confirmation, particularly for complex items that would otherwise demand careful attention. Industry analyses of AI-assisted labelling report that this effect can be severe enough that AI assistance worsens overall quality relative to fully manual labelling when the underlying model's baseline accuracy falls below roughly 80–85%, because a high per-item error rate combined with reviewer over-trust compounds rather than corrects mistakes. This is consistent with the academic literature on annotator drift and fatigue, in which the interpretation of labelling guidelines gradually shifts over the course of a long project, and with evidence that majority voting among multiple human experts does not necessarily resolve disagreement optimally, since no single expert's labels may serve as an unimpeachable ground truth even before AI is introduced. The practical implication is that the accuracy benefit of AI assistance is conditional on model quality, task design, and reviewer behaviour, not a fixed property of "using AI."

3.4 Reducing annotation cost without sacrificing accuracy: active learning

A separate but related technical strand addresses accuracy indirectly, by reducing how much manual annotation is required in the first place. Active learning selects the most informative unlabelled instances for human annotation — using criteria such as prediction uncertainty, expected model change, or query-by-committee disagreement — rather than labelling data at random, so that a model can reach a target level of performance with substantially fewer labelled examples (Settles, 2009). Reported reductions in required annotation volume vary by domain and query strategy, but multiple applied studies describe achieving 95% of full-dataset model performance while annotating only a minority of the available data, for example in retail product recognition using active learning with a Mask R-CNN detector, where annotating roughly 21–24% of the data reached 95% of full-dataset performance. Because active learning changes which items are labelled rather than how they are labelled, it is complementary to, rather than a substitute for, the AI-assistance strategies discussed above: a pipeline can simultaneously use active learning to prioritise which items need human attention and AI-assisted pre-labelling to speed the labelling of each selected item.

Table 1. Summary comparison of manual, fully automated, and hybrid AI-assisted labelling approaches on the dimensions most frequently reported in the reviewed literature.

Dimension	**Manual (expert/crowd) annotation**	**Fully automated / AI-only labelling**	**Hybrid human-in-the-loop (AI-assisted)**
Typical accuracy on well-defined tasks	High, but variable across annotators; inter-rater κ often 0.37–0.90 in systematic-review screening	Comparable to or exceeding crowd workers on structured tasks (e.g., ~25-point gain over MTurk in Gilardi et al.)	Often the highest observed accuracy; e.g., 90–92% after AI-assisted verification vs. 82–87% before
Handling of ambiguous/nuanced items	Strong; captures sarcasm, cultural context, domain nuance	Weaker; struggles with edge cases and rare classes	Improved; model flags uncertain cases for human review
Throughput / scalability	Slow, linear in annotator-hours	Very high; near-constant marginal cost per item	Moderate-high; bottlenecked by human review of flagged items
Cost per label	Highest (skilled labour, training, QA)	Lowest (compute cost only)	Intermediate; lower than pure manual, higher than pure automated

Consistency across large datasets	Subject to fatigue, drift, and inter-annotator disagreement	Highly consistent, but can propagate systematic model bias	Consistency improved through calibration and consensus rules
Key failure mode	Fatigue, subjectivity, drift over long projects	Automation bias risk is inherited by downstream users; hallucination in generative outputs	Automation bias in reviewers who passively confirm AI output

4. Current Approaches and Developments

Several concrete workflows illustrate how AI assistance is currently deployed on scientific data:

- Systematic-review screening. Multiple groups now use LLMs for title-and-abstract screening as an adjunct or filter ahead of human review. Khraisha et al. (2023) evaluated GPT-4 on peer-reviewed and grey literature across multiple languages; the validation study of Llama 3 and ChatGPT-4o mini (PMC12623132) reports sensitivity/specificity trade-offs at the article level; and Chen et al. (2025) implement an explicit human-LLM collaboration protocol for randomized-controlled-trial screening in precision oncology, reporting improved reliability over unassisted GPT-3.5 screening.
- Data extraction from full text. Mahmoudi et al. (2025) critically assess ChatGPT's performance extracting structured data fields from full-text articles for systematic reviews, an area where LLM outputs must be checked against source text because of documented risks of fabricated or inaccurate references and figures, as demonstrated by Chelli et al. (2024) in their analysis of hallucination rates and reference accuracy for ChatGPT and Bard in systematic-review contexts.
- Medical image and figure annotation. Dreizin et al. (2023) apply AI-collaborative labelling to CT trauma volumetry with explicit quality-assurance and bias-mitigation components; Chen, Zhou, and Yuille (2024) propose an automatic label-quality estimator (Quality Sentinel) applicable across 142 anatomical structures to flag both manual and AI-generated segmentation labels of doubtful quality; and Yamagata and Yamada (2024) evaluate GPT-4 with vision for annotating cellular-senescence figures extracted from review articles, reporting over 70% accuracy for label extraction and approximately 80% accuracy for higher-level categorisation.
- Metadata curation for data discovery. Turner et al. (2025) apply GPT-4o to extract structured metadata (scanner parameters, participant characteristics, study design fields) from neuroimaging publications for the NeuroBridge data-discovery project, reporting agreement with human gold-standard annotation in the 0.91–0.97 range on several extraction sub-tasks.
- Weakly and distantly supervised label induction for scientific documents. Rather than using AI to check human labels, some pipelines induce labels automatically from structural cues in scientific documents

themselves; Siegel, Lourie, Power, and Ammar (2018) generate over 5.5 million induced figure-extraction labels from arXiv and PubMed documents with no human annotation step, illustrating an alternative, fully automated route to large labelled scientific corpora where manual labelling at comparable scale would be impractical.

- Cost-reduction infrastructure. Commercial annotation platforms (e.g., Amazon SageMaker Ground Truth, Scale AI, and comparable tools referenced in recent dataset-safety literature for autonomous-driving perception data) now bundle semi-automated, machine-learning-assisted annotation with built-in quality-control mechanisms intended to preserve label accuracy and reliability while reducing per-label human effort, reflecting the same hybrid principle documented in the academic literature above.

5. Research Challenges and Limitations

5.1 Absence of a stable ground truth

A recurring methodological difficulty is that "manual annotation" is frequently treated as ground truth in comparative studies, even though the confident-learning literature demonstrates that manually curated benchmark test sets themselves contain pervasive label errors — at least 3.3% on average and as high as roughly 6% for ImageNet's validation set (Northcutt et al., 2021). Where multiple human experts disagree, majority voting has been shown not to reliably produce the best obtainable consensus, meaning that accuracy figures computed against any single human reference (or an unweighted panel) may misstate the true performance gap between AI-assisted and manual pipelines.

5.2 Heterogeneity of tasks, models, and metrics

Reported accuracy figures are not directly comparable across studies because they span different tasks (text classification, image segmentation, figure annotation, metadata extraction), different models (GPT-3.5, GPT-4, GPT-4o, Llama 3, and others), different prompting strategies (zero-shot versus few-shot versus fine-tuned), and different accuracy metrics (raw accuracy, sensitivity/specificity, Cohen's kappa, intercoder agreement). Delgado-Chaves et al.'s reported 40–92% accuracy range across 18 LLMs and three clinical domains illustrates how much of the apparent "AI versus manual" gap is really a product of model and prompt selection rather than a fixed property of AI assistance as such.

5.3 Domain and language imbalance in the evidence base

Much of the strongest quantitative evidence for AI-assisted annotation accuracy comes from English-language biomedical systematic reviews and general social-media text classification. Studies in other scientific domains — physical sciences, astronomy, materials science, and non-English scientific corpora — are comparatively sparse, and where they exist, such as the Arabic discourse-annotation study by Al-Athba and Zaghouani (2026), results can differ from the English biomedical literature, with manual annotation retaining a clearer advantage on culturally and linguistically nuanced material.

5.4 Automation bias and reviewer behaviour are under-measured

Although automation bias is widely discussed as a risk of AI-assisted labelling pipelines, few of the peer-reviewed studies reviewed here directly measure how reviewer verification behaviour changes when confronted with a pre-filled AI label versus a blank field, as opposed to simply reporting final accuracy after verification. This leaves an important causal question only partially answered: how much of any measured accuracy gain in hybrid pipelines is attributable to the AI system's raw suggestions being correct, versus to the review process itself catching errors, versus being offset by reviewers rubber-stamping incorrect AI suggestions.

5.5 Reproducibility and versioning of proprietary models

Because most of the strongest empirical results rely on proprietary, frequently updated commercial LLMs (ChatGPT/GPT-4-family models), accuracy figures reported at one point in time may not be reproducible with later model versions, and cost figures (e.g., the sub-cent per-annotation cost reported by Gilardi et al., 2023) can shift substantially as pricing and model architectures change, limiting the durability of quantitative benchmarks in this literature.

6. Research Gaps and Future Directions

- Domain-diverse benchmarking. There is a clear need for large-scale, pre-registered comparisons of AI-assisted versus manual labelling accuracy across a wider range of genuinely scientific datasets — physical sciences, environmental and earth sciences, astronomy, chemistry — rather than the current concentration in biomedical text and imaging.
- Causal decomposition of hybrid-pipeline gains. Future work should experimentally separate the contribution of AI pre-labelling quality from the contribution of human verification behaviour (including automation bias), for example via randomized designs that vary whether reviewers see AI-suggested labels, blinded alternative labels, or no suggestion at all.
- Better-than-majority-vote ground truth construction. Building on findings that majority voting among experts is not always optimal, further methodological work is needed on principled ways to construct reference labels — for example weighted consensus, expertise-adjusted aggregation, or confident-learning-style joint-distribution estimation — against which both manual and AI-assisted pipelines can be fairly evaluated.
- Standardised cost-accuracy reporting. Because cost and accuracy trade off differently across tasks, the field would benefit from standardised reporting of accuracy at matched cost (or cost at matched accuracy), rather than accuracy and cost reported as separate, non-comparable headline numbers.
- Integrating active learning with AI-assisted verification. Relatively little published work directly evaluates pipelines that combine active-learning sample selection with AI-assisted pre-labelling and human verification as a single system, despite each component individually showing accuracy or efficiency benefits.
- Longitudinal and versioning-aware evaluation. Given the pace of LLM development, future studies should report results in a way that is explicit about model version and date, and ideally re-run key benchmarks

periodically, to track whether the AI-assisted accuracy advantage over manual and crowd-based annotation is growing, stable, or task-dependent over time.

- Non-English and low-resource scientific corpora. Extending accuracy comparisons to non-English scientific literature and to low-resource languages is an open gap, given evidence that current LLMs may underperform on culturally and linguistically nuanced annotation relative to trained human annotators in those settings.

7. Conclusion

The evidence reviewed here does not support a simple claim that AI-assisted labelling is categorically more or less accurate than manual labelling on scientific datasets. Instead, accuracy depends on task structure, model choice, and — critically — pipeline design. On well-defined classification and extraction tasks, AI assistance frequently matches or exceeds the accuracy of non-expert crowd annotation and, in several biomedical and neuroimaging studies, approaches the accuracy of trained human annotators. On tasks requiring deep contextual or cultural judgement, unaided manual annotation retains a measurable advantage. The most robust and consistently reported finding across the literature is that hybrid, human-in-the-loop pipelines — in which AI systems generate candidate labels that trained humans verify and correct — tend to outperform either pure approach in isolation, provided that automation bias is actively managed and the underlying model's baseline accuracy is sufficient to make verification worthwhile. Because manually curated benchmark datasets themselves contain non-trivial label error, and because the evidence base remains concentrated in English-language biomedical and text-classification tasks, substantial further work is needed — spanning broader scientific domains, more rigorous decomposition of where hybrid-pipeline accuracy gains come from, and standardised, cost-aware accuracy reporting — before general conclusions about AI-assisted versus manual labelling accuracy can be drawn with confidence across the sciences as a whole.

References

Al-Athba, A. A., & Zaghouani, W. (2026). Cohesion-6K: An Arabic dataset for analyzing social cohesion and conflict in online discourse. arXiv. https://arxiv.org/abs/2605.22447 (arXiv:2605.22447)

Chelli, M., Descamps, J., Lavoué, V., Trojani, C., Azar, M., Deckert, M., Raynier, J.-L., Clowez, G., Boileau, P., & Ruetsch-Chelli, C. (2024). Hallucination rates and reference accuracy of ChatGPT and Bard for systematic reviews: Comparative analysis. Journal of Medical Internet Research, 26, e53164. https://doi.org/10.2196/53164

Chen, H., Zhao, J., Zheng, S., Zhang, X., Duan, H., & Lu, X. (2025). A human-LLM collaborative annotation approach for screening articles on precision oncology randomized controlled trials. BMC Medical Research Methodology, 25. https://doi.org/10.1186/s12874-025-02674-3

Chen, Y., Zhou, Z., & Yuille, A. (2024). Quality Sentinel: Estimating label quality and errors in medical segmentation datasets. arXiv. https://arxiv.org/abs/2406.00327 (arXiv:2406.00327)

Delgado-Chaves, F. M., et al. (as cited in: Large language models in systematic review screening: Opportunities, challenges, and methodological considerations). (2025). Information, 16(5), 378. https://doi.org/10.3390/info16050378

Dreizin, D., Zhang, L., Sarkar, N., Bodanapally, U. K., Li, G., Hu, J., Chen, H., Khedr, M., Khetan, U., Campbell, P., & Unberath, M. (2023). Accelerating voxelwise annotation of cross-sectional imaging through AI collaborative labeling with quality assurance and bias mitigation. Frontiers in Radiology, 3, 1202412. https://doi.org/10.3389/fradi.2023.1202412 (PMCID: PMC10362988)

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. Proceedings of the National Academy of Sciences of the United States of America, 120(30), e2305016120. https://doi.org/10.1073/pnas.2305016120 (PMID: 37463210)

Khraisha, O., Put, S., Kappenberg, J., Warraitch, A., & Hadfield, K. (2023). Can large language models replace humans in the systematic review process? Evaluating GPT-4's efficacy in screening and extracting data from peer-reviewed and grey literature in multiple languages. arXiv. https://arxiv.org/abs/2310.17526 (arXiv:2310.17526)

Mahmoudi, H., Chang, D., Lee, H., Ghaffarzadegan, N., & Jalali, M. S. (2025). Critical assessment of large language models' (ChatGPT) performance in data extraction for systematic reviews: Exploratory study. JMIR Publications. https://doi.org/10.2196/68097 (PMCID: PMC12425462)

Northcutt, C. G., Athalye, A., & Mueller, J. (2021). Pervasive label errors in test sets destabilize machine learning benchmarks. Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS 2021), Datasets and Benchmarks Track. arXiv. https://arxiv.org/abs/2103.14749 (arXiv:2103.14749)

Northcutt, C. G., Jiang, L., & Chuang, I. L. (2021). Confident learning: Estimating uncertainty in dataset labels. Journal of Artificial Intelligence Research, 70, 1373–1411. https://doi.org/10.1613/jair.1.12125 (arXiv:1911.00068)

Settles, B. (2009). Active learning literature survey (Computer Sciences Technical Report 1648). University of Wisconsin–Madison.

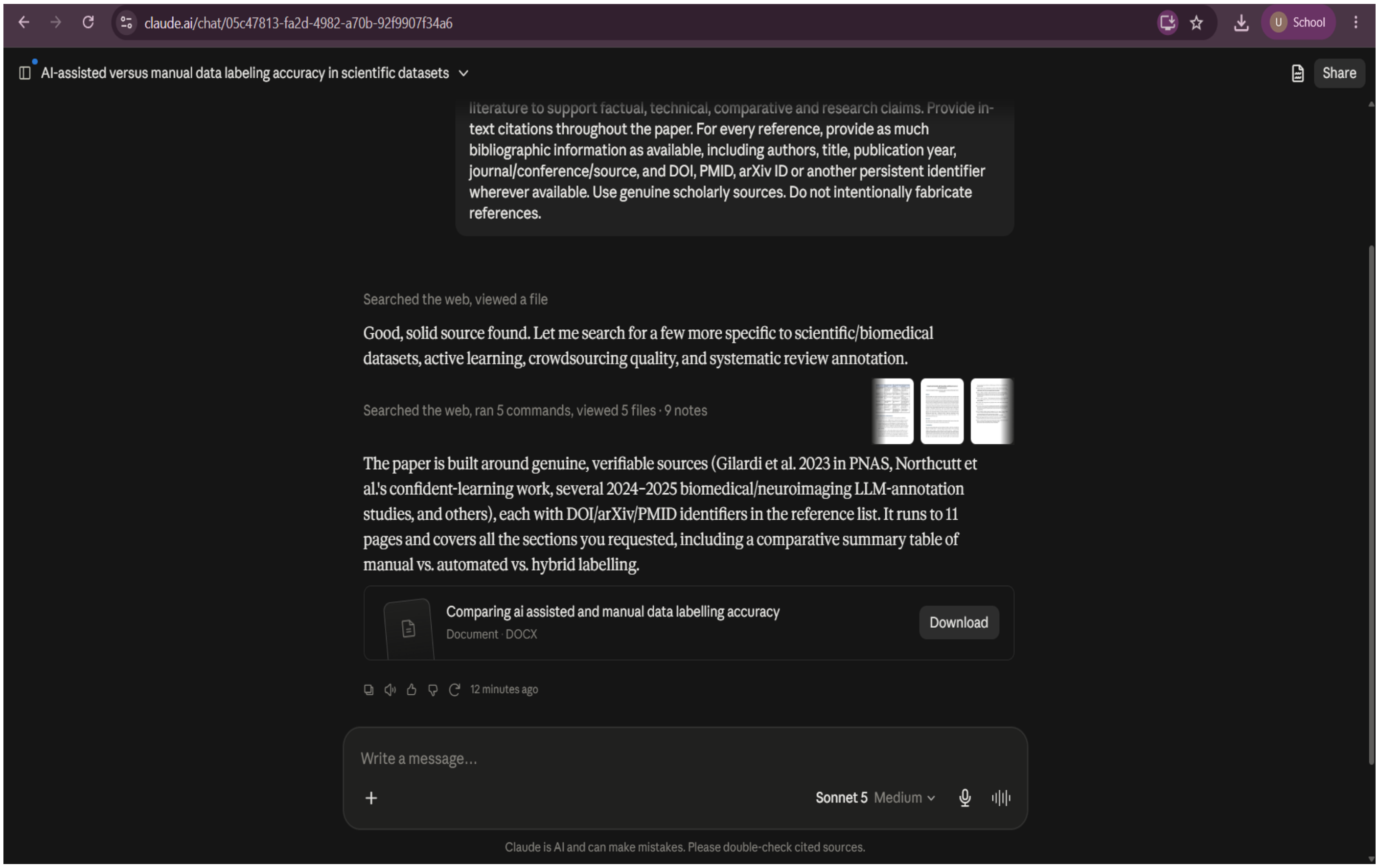
Siegel, N., Lourie, N., Power, R., & Ammar, W. (2018). Extracting scientific figures with distantly supervised neural networks. In Proceedings of the 18th ACM/IEEE Joint Conference on Digital Libraries (JCDL '18) (pp. 223–232). Association for Computing Machinery. https://doi.org/10.1145/3197026.3197040 (arXiv:1804.02445)

Turner, M. D., Appaji, A., Ar Rakib, N., Golnari, P., Rajasekar, A. K., K V, A. R., Sahoo, S. S., Wang, Y., Wang, L., & Turner, J. A. (2025). Large language models can extract metadata for annotation of human neuroimaging publications. Frontiers in Neuroinformatics, 19, 1609077. https://doi.org/10.3389/fninf.2025.1609077 (PMID: 40909182; PMCID: PMC12405296)

Validation of large language models (Llama 3 and ChatGPT-4o mini) for title and abstract screening in biomedical systematic reviews. (2025). [Journal article]. PMCID: PMC12623132.

Yamagata, Y., & Yamada, R. (2024). Survey on large language model annotation of cellular senescence from figures in review articles. Discover Applied Sciences / related journal, 2024. https://doi.org/10.1186/s44342-024-00011-6 (PMCID: PMC11800539)

Claude ScreenShot:



Claude chat Link:

https://claude.ai/share/30137af5-df18-4e8e-bd93-503383434413